Advanced information, computation, communication I
EPFL - Fall semester 2025-2026

Week 6

October 17, 2025

Exercise 1. (Repeated from Week 5) Solve the following recurrence relation using generating functions: $a_n = 4a_{n-1} - 4a_{n-2}$, $n \geq 2$, with $a_0 = 1$ and $a_1 = 4$.

Exercise 2. Show that $\sum_{j=1}^{n}(a_j - a_{j-1}) = a_n - a_0$, where $a_0, a_1, ..., a_n$ is a sequence of real numbers. This type of sum is called **telescoping**.

Exercise 3. Sum both sides of the identity $k^2 - (k-1)^2 = 2k - 1$ from $k = 1$ to $k = n$ and use Exercise 2 to find

(a) a formula for $\sum_{k=1}^{n}(2k-1)$ (the sum of the first n odd natural numbers).

(b) a formula for $\sum_{k=1}^{n} k$.

Exercise 4. The following three algorithms sort the input sequence $a_1, \ldots, a_n$ of real numbers in ascending order.

Algorithm 1 Bubble Sort

```
for i = 1 to n − 1 do
  for j = 1 to n − i do
    if a_j > a_{j+1} then
      swap a_j and a_{j+1}
    end if
  end for
end for
```

Algorithm 2 Selection Sort

```
for i = 1 to n − 1 do
  min ← i + 1
  for j = i + 1 to n do
    if a_min > a_j then
      min ← j
    end if
  end for
  if a_i > a_min then
    swap a_i and a_min
  end if
end for
```

Algorithm 3 Insertion Sort

```
for j = 2 to n do
  i ← 1
  while a_j > a_i do
    i ← i + 1
  end while
  m ← a_j
  for k = 0 to j − i − 1 do
    a_{j−k} ← a_{j−k−1}
  end for
  a_i ← m
end for
```

Use Bubble Sort, Selection Sort and Insertion Sort to sort the following sequence:

`9, 12,-43, 20, -2, 3, 7, 28, 19.`

Exercise 5. Adapt the bubble sort algorithm so that it stops when no swaps are required (when the list is sorted). Express this more efficient version of the algorithm in pseudocode.

Exercise 6. Two strings are anagrams if each can be formed from the other string by rearranging its characters. Devise an algorithm to determine whether two strings are anagrams

Exercise 7. Describe an algorithm that determines whether a function from a finite set to another finite set is injective.

Exercise 8. We use Bubble Sort, Selection Sort, and Insertion Sort to sort the following sequence (see Exercise 3):

`9, 12,-43, 20, -2, 3, 7, 28, 19.`

1. How many comparisons are done in each of the algorithms?
2. How many swaps are done in each of the algorithms?
3. In the worst-case scenario, what is the order of how many comparisons and how many swaps are done by the three algorithms for an input sequence of length $n + 1$?